

Manas Ganti

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M.S. Computer Engineering candidate | RL environments, verifiable rewards, VLM post-training, and evaluation systems

Education

Virginia Polytechnic Institute and State University (Virginia Tech)

MASTER OF SCIENCE IN COMPUTER ENGINEERING

Blacksburg, Virginia
Aug. 2025 – Expected May 2027

MIT World Peace University (MIT WPU)

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

Pune, India
Jun. 2018 – May 2022

Technical Skills

Reinforcement Learning	RL environment design, verifiable rewards (RLVR), reward modeling and shaping, reward-hacking analysis, GRPO, PPO, SAC, RLHF, Gymnasium, PettingZoo
Post-Training & LLMs	SFT, LoRA/PEFT, distillation, rejection sampling, TRL, Transformers, vLLM, Qwen2.5, Qwen2.5-VL, tool-using agents, LangGraph, RAG
Training & Evaluation	DeepSpeed ZeRO-2/3, SLURM, A100/H200 clusters, tensor-parallel inference, throughput and memory profiling, evaluation harnesses, verifier construction, ablations, calibration, confound and leakage detection, Weights & Biases, CI-tested reward logic
Vision & Systems	PyTorch, computer vision, VLMs, DeepStream, TensorRT, ONNX, quantization, edge deployment, Python, C++, SQL, Bash, FastAPI, PostgreSQL/pgvector, Docker, AWS, GCP, Git

Selected Projects

Investigative Visual Reasoning via Verifiable Reinforcement Learning [GitHub](#)

QWEN2.5-VL · PYTORCH · TRL · VLLM · DEEPSPEED · SLURM

- Built a partially observable VLM environment with INSPECT and VERDICT actions, requiring a policy to acquire high-resolution visual evidence under a limited inspection budget before a terminal real-versus-synthetic image decision.
- Implemented deterministic, unit-tested trajectory rewards for **label correctness**, calibrated confidence, evidence consistency, and inspection cost; rewards use ground-truth labels and structured policy fields without an LLM judge.
- Constructed a 1,570-image geometry-matched paired benchmark and removed a dimension-based shortcut, reducing label predictability from **0.85 to 0.50 AUC**; built degradation/budget-shift evaluations, ceiling probes, reward-gaming tests, and a **137-test CI suite**.
- Implemented LoRA SFT→GRPO post-training and scalable A100/H200 workflows using DeepSpeed ZeRO-2/3, vLLM tensor-parallel evaluation, SLURM job chaining, and Weights & Biases experiment tracking.

Long-Context Navigation Environment [GitHub](#)

PYTHON · PYTORCH · PEFT · VLLM · SLURM

- Built a verifiable multi-hop RL environment for **33K–200K-token** documents under a strict **600-token working-memory ceiling**, with READ, COMPRESS, DROP, and ANSWER actions and exact token accounting.
- Generated procedurally validated tasks with adversarial near-miss distractors and population-level leakage audits; eliminated a size-biased sampling artifact, reducing label prediction from **0.60 to 0.50 AUC** before training.
- Implemented SFT→GRPO on NVIDIA A100 GPUs and instrumented training to diagnose incorrect KL references, length-biased gradients, and unsynchronized ranks; added probes for fact retention and redundant re-read loops.

Clinical Intelligence Platform: Production Full-Stack AI Data System [GitHub](#)

PYTHON · FASTAPI · PYDANTIC V2 · POSTGRES SQL · PGVECTOR · REACT · TYPESCRIPT · DOCKER

- Built and shipped a full-stack clinical dashboard using **FastAPI**, **PostgreSQL/pgvector**, **React/TypeScript**, and Docker; ingested **1,171 patients** and **150K laboratory findings** through typed Pydantic v2 pipelines with schema-coercion adapters, provenance, and reference-range alerts.
- Engineered a constrained LLM query layer using Claude function calling and a **32-case evaluation harness** measuring tool-call correctness across six task categories.
- Added server-side pagination and time-bucketed chart aggregation; reduced a high-volume LLM tool call from **58K to 2.9K tokens** by pushing filtering operations into SQL.

Work Experience

Assert AI

May 2024 – Mar. 2025

ASSOCIATE AI ENGINEER · Python, NumPy, PyTorch, DeepStream, Computer Vision

- Implemented real-time computer-vision pipelines using NVIDIA **DeepStream** on Jetson Nano and Orin AGX devices, optimizing latency, GPU utilization, and throughput for industrial safety monitoring across multi-camera deployments.
- Fine-tuned and deployed **YOLOv8** pose-estimation models for hazardous-zone detection, reducing false alerts from **14% to under 3%** on 8,400 held-out frames while tuning the precision–recall operating point for asymmetric safety costs.
- Engineered a production **ANPR** system processing **2,300 vehicles daily** across six facilities, replacing manual gate logging and reducing compliance-review effort by **70%**.
- Deployed custom YOLOv8 PPE-compliance models across 32 cameras at four sites, achieving **98% recall** with under **2% false-positive rate** on 11,000 held-out day/night, motion-blur, and occlusion-heavy frames.
- Built observability tooling for per-camera detection rates, drift, and alert volumes, surfacing degradation before manual review and informing retraining decisions.

Inmed Prognostics

Oct. 2022 – May 2024

AI / DEEP LEARNING ENGINEER · Python, NumPy, Pandas, OpenCV, NiBabel, ANTsPy, SimpleITK, U-Net, ResNet

- Developed U-Net models with ResNet-34 backbones for brain-lobe segmentation across 250 MRI volumes, achieving **0.89 mean Dice** and **3.2 mm HD95** in 3-fold cross-validation.
- Engineered a patch-based brain-CT hemorrhage pipeline with ResNet-34, improving sensitivity from **81% to 93%** on 260 held-out scans while reducing inference latency from **18 seconds to 6 seconds** per volume.
- Built scalable preprocessing pipelines for Z-score normalization, 2D slice extraction, intensity standardization, and augmentation to improve reproducibility across heterogeneous clinical-imaging datasets.
- Collaborated with frontend and backend engineers to ship four deep-learning models into clinical-imaging systems, designing REST inference APIs serving approximately 100 scans per day.